

Physical Layer

The physical layer is the first layer of the OSI (Open Systems Interconnection) model in computer networking. It deals with the transmission of raw data over physical media like cables, fiber optics, or wireless signals. It defines the hardware elements involved in the communication, such as the electrical signals, radio waves, and connectors. In short, the physical layer is responsible for the actual transmission of bits across a medium.

In the physical layer, the transmission of data is represented as a series of **0s and 1s**, known as **bits**. These bits are encoded into signals (electrical pulses, light signals, or radio waves) that can travel through the physical medium (like copper wires, fiber optics, or air for wireless communication).

Signal

A **signal** is a **physical representation of data** used to transmit information from one device to another in a computer network.

It is a **time-varying electrical, electromagnetic, or optical wave** that carries data through a transmission medium such as a wire, fiber cable, or air.

In simple words, **signals are the form in which data travels in a network**.

To be transmitted, data must be transformed to electromagnetic signals. Data can be analog or digital.

Analog data refers to information that is continuous. Analog data, such as the sounds made by a human voice, take on continuous values.

Digital data refers to information that has discrete states. Digital data take on discrete values. **For example, data are stored in computer memory in the form of 0s and 1s.**

They can be converted to a digital signal or modulated into an analog signal for transmission across a medium.

A **signal** is a **way to send data** from one computer to another.

When a computer wants to send data, it **changes the data into signals**. These signals **travel through wire, cable, or air**.

The receiving computer **converts the signal back into data**.

Easy example

When you press **A** on the keyboard:

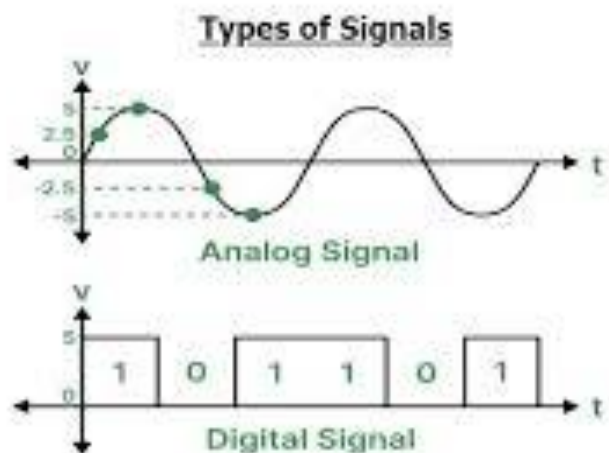
- Computer changes **A** into **0 and 1**
- These **0 and 1 travel as signals**
- Other computer receives them and shows **A**

Without signals, computers cannot communicate.

On the Basis of Type

Signals can be classified on the basis of its type into following two groups:

- 1. Analog Signal: They have infinite values in a range.**
- 2. Digital Signal: They have limited number of defined values**



Periodic & Non-Periodic Signals:

Signals which repeat itself after a fixed time period are called Periodic Signals.

Signals which do not repeat itself after a fixed time period are called Non-Periodic Signals.

In data communications, we commonly use periodic analog signals and non-periodic digital signals.

Analog Signal

Signals can be either analog or digital. An analog signal has infinitely many levels of intensity over a period of time. As the wave moves from value A to value B, it passes through and includes an infinite number of values along its path.

- Analog signals vary smoothly over time.
- There are no sudden jumps between values.

An analog signal is a signal that is continuous in time and continuous in amplitude. This means it can take any value at any instant of time within a given range.

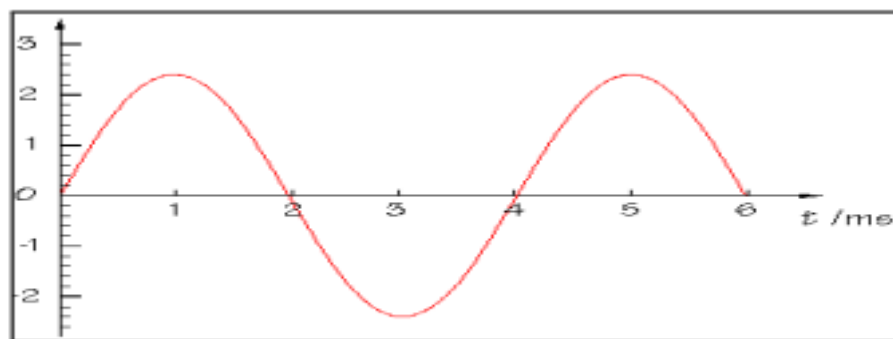


Fig. Sine wave

Sine Wave

The sine wave is the most fundamental form of a periodic analog signal.

Smooth wave that **starts from 0**.

Goes **up and down repeatedly**.

Cosine Wave

Smooth wave that **starts from maximum value**

Also goes **up and down repeatedly**.

A sine wave is characterized by three parameters:

1. Peak Amplitude
2. Frequency
3. Phase

Peak Amplitude

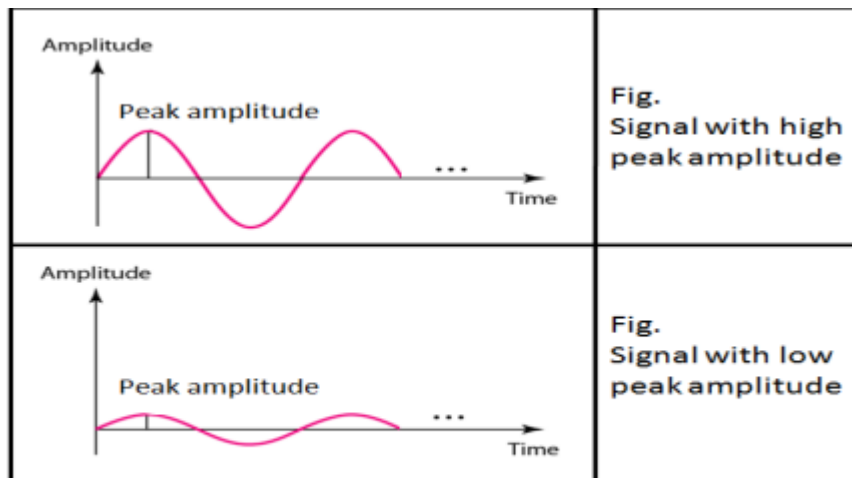
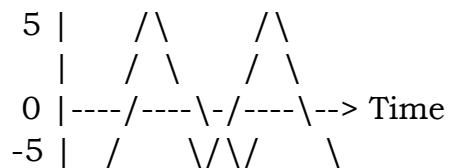


Fig. Amplitude of a sine wave

The **peak amplitude** is **the highest point of a wave** above or below zero.
The amplitude of a signal is proportional to the energy carried by the signal.

Example:

If a sine wave goes up to +5 volts and down to -5 volts,
The peak amplitude = 5 volts
Amplitude



The top of the wave = +5 → peak amplitude

The bottom of the wave = -5 → negative peak, still counted as 5

- **Units:** It is commonly measured in **volts (V)** for electrical signals or **decibels (dB)** for sound waves.

Period & Frequency

Period refers to the amount of time, in seconds, a signal needs to complete 1 cycle.

Unit- Period is formally expressed in seconds.

Frequency:

Frequency is the number of times a wave repeats in one second.

Frequency refers to the number of periods in 1s.

Unit- Frequency is formally expressed in Hertz (Hz), which is cycle per second.

- Measured in Hertz (Hz).
- 1 Hz = 1 wave per second

Frequency is the rate of change with respect to time. Change in a short span of time means high frequency. Change over a long span of time means low frequency.

Example:

- If a wave repeats 5 times in 1 second, its frequency = 5 Hz
- AC electricity in homes (India) has 50 Hz, meaning the wave repeats 50 times per second.

Frequency is the number of waves passing a point in one second.

Period is the inverse of frequency, and frequency is the inverse of period, as the following formulas show.

Relation between Frequency & Period Frequency & Period are inverse of each other.

It is indicated by the following formula:

$$\begin{array}{c} T = 1/f \\ \text{Or} \\ f = 1/T \end{array}$$

Example1. A wave has a frequency of 100hz. Its period(T) is given by
 $T = 1 / F = 1 / 100 = 0.01 \text{ sec}$

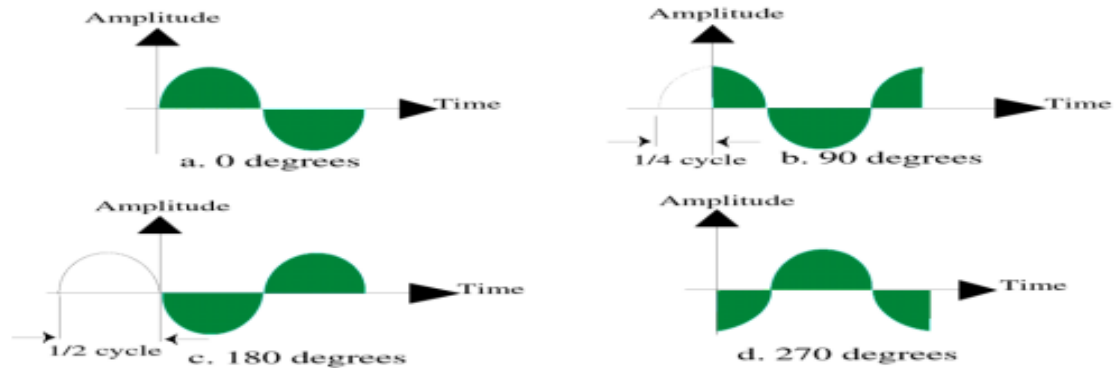
Example2. A wave completes its one cycle in 0.25 seconds. Its frequency is given by

$$F = 1 / T = 1 / 0.25 = 4 \text{ Hz}$$

Phase

The term phase describes the position of the waveform relative to time 0. If we think of the wave as something that can be shifted backward or forward along the time axis, phase describes the amount of that shift. It indicates the status of the first cycle.

Phase of a wave is the **horizontal shift of the wave from its usual starting point**. It shows **where the wave begins** compared to a reference wave.



Unit

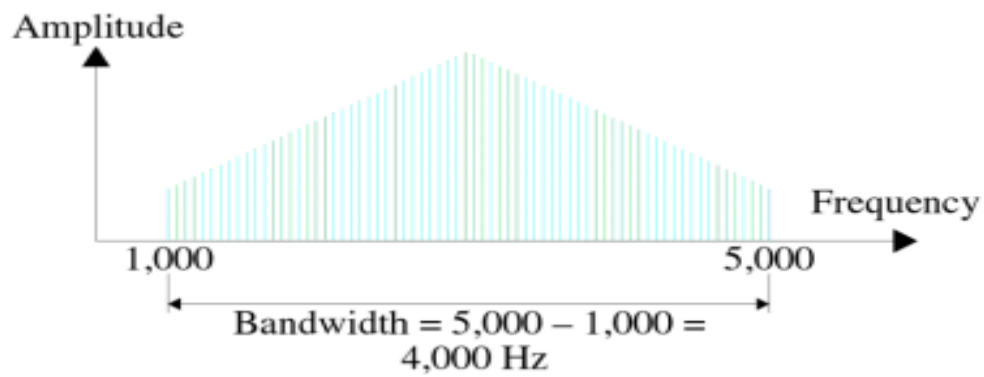
- Degrees ($^{\circ}$) $\rightarrow 360^{\circ} = 1$ full wave

Example

- A normal sine wave starts from 0.
- If the wave starts from its **maximum value** instead of 0, the wave is said to have a **phase of 90°** .

Bandwidth

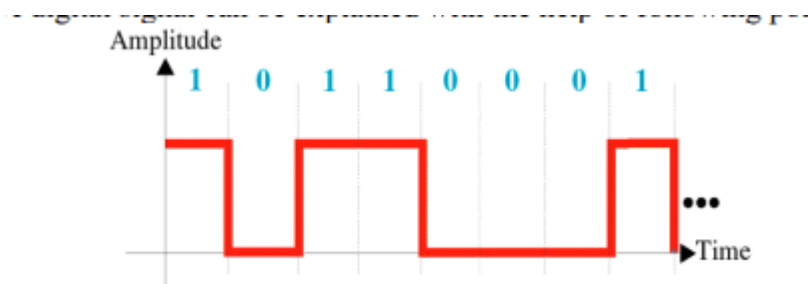
The range of frequencies contained in a composite signal is its bandwidth. The bandwidth is normally a difference between two numbers. For example, if a composite signal contains frequencies between 1000 and 5000, its bandwidth is 5000- 1000, or 4000.



The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

Digital Signals:-

- Information can also be explained in the form of a digital signal.
- A digital is a signal that has discrete values. The signal will have value that is not continuous.
- In addition to being represented by an analog signal, information can also be represented by a digital signal.
- For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels.
- i.e. A digital signal, can have only a limited number of defined values. Although each value can be any number, it is often as simple as 1 and 0.



Information in a digital signal can be represented in the form of voltage levels. Ex. In the signal shown below, a „1‘ is represented by a positive voltage and a „0‘ is represented by a Zero voltage.

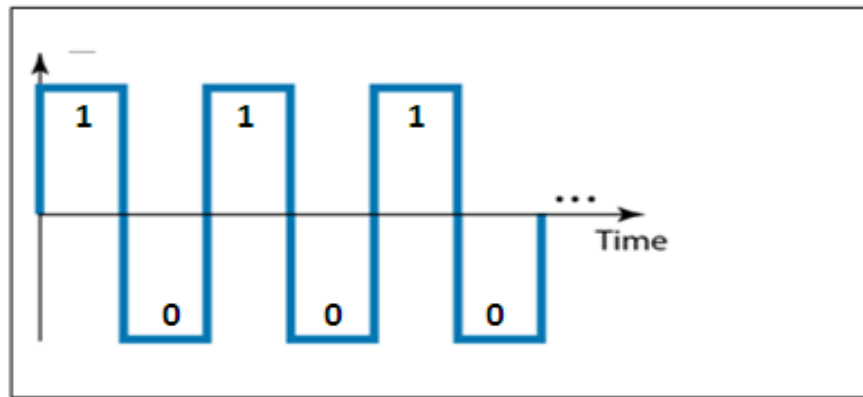


Fig: A digital signal with Two levels. „1“ represented by a positive voltage and „0“ represented by a negative voltage

Bit Rate

Most digital signals are non-periodic, to describe digital signals term bit rate (instead of frequency) is used.

The **bit rate** is the number of bits sent in 1s, expressed in bits per second (bps).

BIT RATE It is the number of bits transmitted in one second. It is expressed as bits per second (bps). Relation between bit rate and bit interval can be as follows

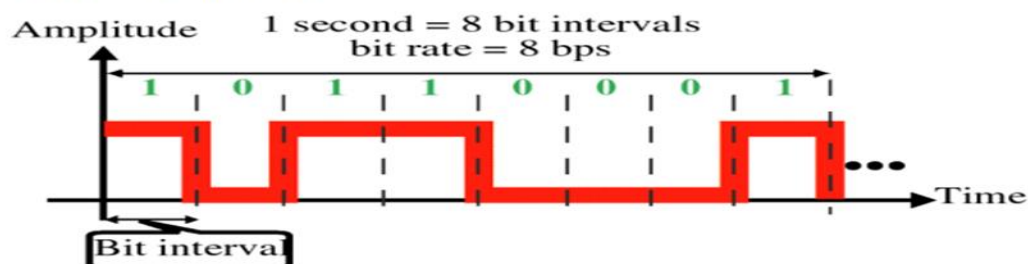
$$\text{Bit rate} = 1 / \text{Bit interval}$$

Bit Length or Bit Interval (T_b)

The bit length is the distance one bit occupies on the transmission medium.

$$\text{Bit length} = \text{propagation speed} \times \text{bit duration}$$

The bit interval is the time required to send one single bit. The bit rate is the number of bit intervals per second.



Bit rate is the number of bits transmitted per second in a digital communication system.

Unit

- bits per second (bps)

Formula

$$\text{Bit Rate}(R) = \frac{\text{Number of bits}}{\text{Time in seconds}}$$

Example

- If 1000 bits are sent in 1 second, then

$$R = \frac{1000}{1} = 1000 \text{ bps}$$

Bit length (or bit duration) is the time taken to transmit one bit.

Unit

- Seconds (s) per bit

Formula

$$\text{Bit Length}(T_b) = \frac{1}{\text{Bit Rate}}$$

Example

- If bit rate $R = 1000$ bps, then

$$T_b = \frac{1}{1000} = 0.001 \text{ seconds per bit} = 1 \text{ ms}$$

FOURIER ANALYSIS

Fourier analysis is a tool that changes a time-domain signal to a frequency-domain signal and vice versa.

Fourier Series

Fourier proved that a composite periodic signal with period T (frequency f) can be decomposed into a series of sine and cosine functions in which each function is an integral harmonic of the fundamental frequency f of the composite signal. The result is called the Fourier series.

$$F(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} a_n \cos(2\pi nft)$$

- $f(t)$
The original periodic signal as a function of time.
- a_0
The DC component (average value) of the signal.
It represents the constant part of the signal.
- $\sum_{n=1}^{\infty}$
Summation from $n = 1$ to infinity, meaning the signal is built by adding infinite sinusoidal components.
- n
The harmonic number (1, 2, 3, ...).
Each value of n represents a frequency that is a multiple of the fundamental frequency.
- a_n
Amplitude of the cosine component of the n -th harmonic.
- b_n
Amplitude of the sine component of the n -th harmonic.
- $\cos(2\pi nft)$
Cosine wave of the n -th harmonic frequency.
- $\sin(2\pi nft)$
Sine wave of the n -th harmonic frequency.
- f
Fundamental frequency of the signal (cycles per second).
- t
Time variable (in seconds).
- 2π
Converts frequency from cycles to radians, since trigonometric functions use radians.

Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment.

When a signal travels from sender to receiver, it does **not arrive perfectly**. This problem is called **transmission impairment**.

Transmission impairment is the reduction in quality of a signal as it travels from the sender to the receiver.

Three causes of impairment are attenuation, distortion, and noise.

Types of Transmission Impairment

1 Attenuation

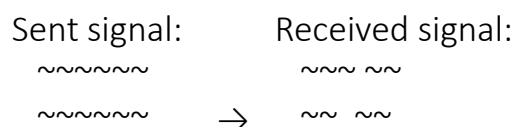


Definition: Loss of signal strength over distance.

Explanation: As a signal travels through a medium (like a cable or air), it becomes weaker. Attenuation means a loss of energy. To compensate for this loss, amplifiers are used to amplify the signal.

Example: A Wi-Fi signal getting weaker as you move farther from the router.

2. Distortion

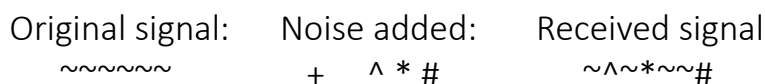


Definition: Change in the shape or form of the signal.

Explanation: Different parts of the signal travel at different speeds, causing the signal to arrive in an altered form. Distortion means that the signal changes its form or shape. Distortion can occur in a composite signal made of different frequencies. signal components at the receiver have phases different from what they had at the sender. The shape of the composite signal is therefore not the same.

Example: A voice sounding unclear during a phone call.

3. Noise



Definition: Unwanted signals that interfere with the original signal.

Explanation: Noise is added to the signal from external sources like electrical devices or natural causes. Noise is another cause of impairment. Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.

Example: Static or crackling sound on a radio.

Note- Data Rate Limits

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

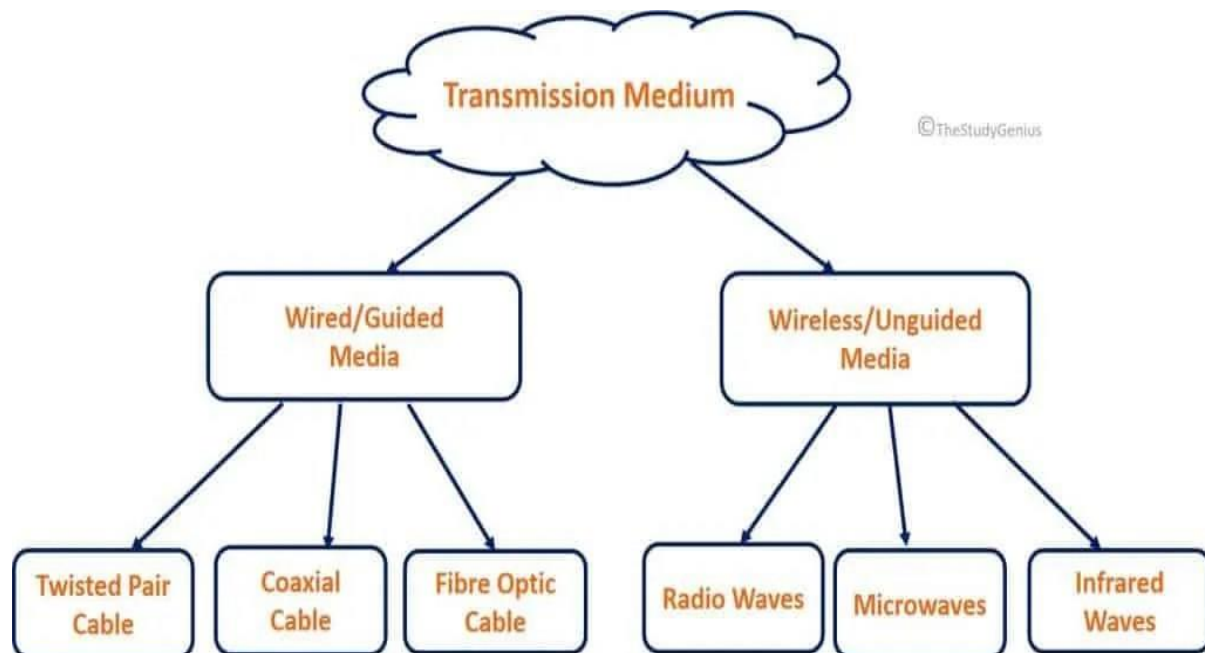
Two theoretical formulas were developed to calculate the data rate:

- 1) by Nyquist for a noiseless channel.

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

- 2) by Shannon for a noisy channel.

$$\text{Capacity} = \text{bandwidth} \times \log_2 (1 + \text{SNR})$$



Transmission Media refers to the physical path or channel through which data signals travel from a sender to a receiver in a communication system.

Types of Transmission Media

1. Guided Media (Wired)

Signals travel through a physical conductor. **Guided media**, also known as **wired** transmission media, refers to physical cables or wires that guide the data signals from one point to another. These physical paths ensure that the signal stays contained within the cable, thus providing a controlled and reliable medium for data transfer.

1. Twisted Pair Cable

Description:

A twisted pair cable consists of two insulated copper wires twisted around each other. The wires help to cancel out electromagnetic interference from external sources by twisting them together.



Insulation- Protect Wires

Copper wire-It carry electrical signals

Twisted Copper Wires-Reduce noise and crosstalk

Types:

- Unshielded Twisted Pair (UTP): No additional shielding; used in most Ethernet networks.
- Shielded Twisted Pair (STP): Has shielding around the wires to further reduce interference.

Usage: Telephone lines, LAN (Local Area Networks), data transmission for low-cost networks.

Advantages

- Low cost
- Easy to install and maintain
- Flexible and lightweight

Disadvantages:

- Limited bandwidth
- Shorter transmission distance
- Easily affected by to electromagnetic interference.

2.Coaxial Cable

A coaxial cable is a type of guided transmission medium in which a central conductor is surrounded by an insulating layer, a metallic shield, and an outer protective layer. The coaxial design ensures that the signal is Enclosed to the central conductor and protected from external electromagnetic interference. Coaxial cables are widely used for high-frequency and long-distance data transmission.

A coaxial cable is a transmission medium in which a central copper conductor is surrounded by insulation and a metallic shield to carry electrical signals with minimal interference and signal loss.

Usage: Cable TV, broadband internet, and some network installations.

Structure of Coaxial Cable

1. Inner Conductor – Usually made of copper, carries the signal.
2. Insulation Layer – Surrounds the conductor to prevent direct contact with the shield.
3. Metallic Shield – Made of Twisted copper or aluminum, prevents electromagnetic interference.
- 4.Outer Jacket – Protective plastic cover to provide mechanical protection.

Types of Coaxial Cable

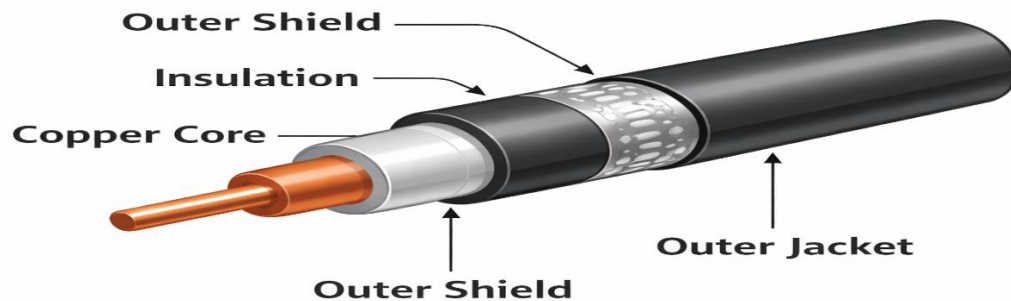
1. Thick Coaxial Cable (10Base5)
 - o High bandwidth and long-distance support
 - o Less flexible, difficult to install
2. Thin Coaxial Cable (10Base2)
 - o Lightweight and flexible
 - o Easier to install, but supports shorter distances

Advantages

- High bandwidth compared to twisted pair
- Better noise and interference immunity
- Suitable for long-distance transmission

Disadvantages

- More expensive than twisted pair
- Bulkier and less flexible
- Difficult to Join or install in tight spaces



3. Optical Fiber Cable

Optical Fiber Cable is a **guided transmission medium** that uses **light signals** to transfer data instead of electricity.

How it works

- Data is converted into **light**
- Light travels through **thin glass or plastic fibers**
- Light reaches the receiver with **very high speed**

Types:

- Single-Mode Fiber: For long-distance communication, where the light follows a single path.
- Multi-Mode Fiber: For shorter distances, where the light can bounce multiple times within the core.

Usage: Internet backbones, high-speed data communication, and long-distance transmission.

Advantages

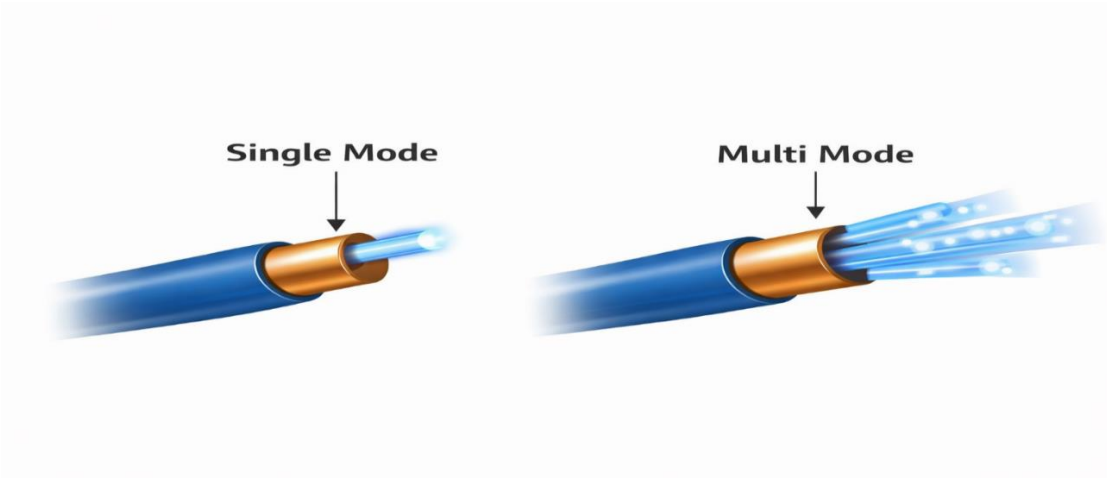
- Very high speed
- Long distance transmission
- No electromagnetic interference
- High security

Disadvantages

- Expensive
- Difficult to install and repair

Optical fiber is the **fastest and safest** way to transmit data.

Core – Thin glass or plastic center through which light travels.



2. Unguided Media (Wireless)

Signals travel through air or space without physical conductors.

Unguided Media (Wireless) refers to data transmission without physical cables, where signals travel through the air (or space) using electromagnetic waves.

Sending data without wires

Instead of using cables, the data **travels through the air**.

- **Phone call** → no wire → your voice goes through the air
- **Wi-Fi** → internet comes to your phone through the air
- **Bluetooth** → music goes through the air to your headphones

All of these are **unguided media**.

Key characteristics

- No physical medium (no wires or fiber)
- Uses antennas for transmission and reception
- Supports mobility and wide coverage

Advantages

- Easy and fast to deploy
- Supports mobile devices
- Covers large or remote areas

Disadvantages

- More prone to interference and noise
- Generally, less secure than guided media
- Can be affected by weather and obstacles

1. Radio Waves

Radio waves are a type of unguided transmission medium that transmit data using electromagnetic waves in the low to medium frequency range.

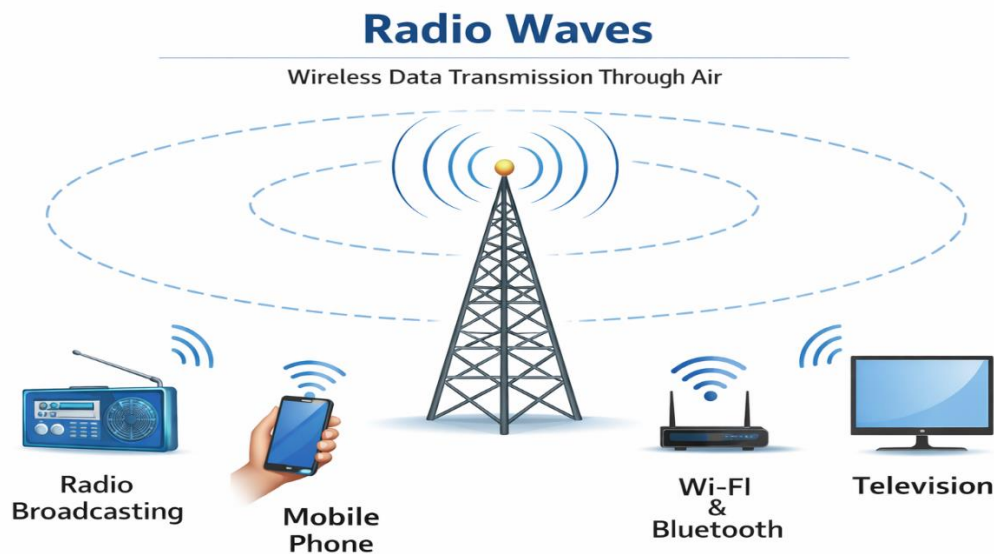
Radio waves are electromagnetic waves with low frequency and long wavelength. They can travel very far and pass through walls and buildings.

Radio waves are a type of **unguided (wireless) transmission media** used to send data **through air** without cables.

Radio waves are **electromagnetic waves**

They travel in **all directions**

No physical medium (wire) is needed



Electromagnetic Waves for Wireless Communication

How they work

- A transmitter converts data (voice, music, internet) into radio signals
- The signal travels through the air
- A receiver (phone, radio, router) receives and converts it back to data

Characteristics

- Long distance communication
- Can pass through walls and buildings
- Low cost
- Easy to install

Uses

- Wi-Fi (home & office internet)
- Bluetooth (wireless mouse, headphones)
- FM/AM radio broadcasting
- Walkie-talkies

Advantages

- Long-distance communication
- Can penetrate obstacles
- Cheap and easy to use

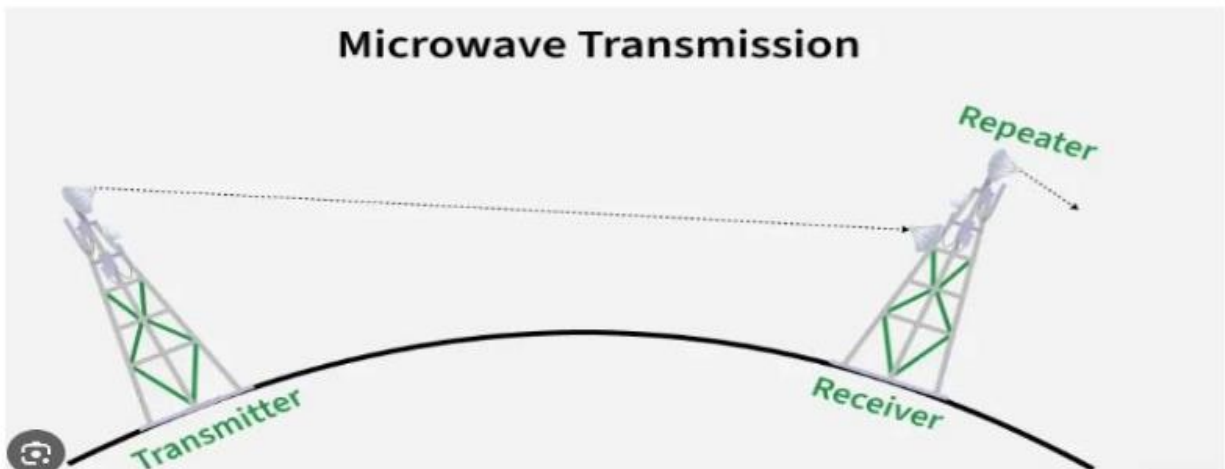
Limitations

- Slower data speed compared to microwaves
- More interference from other signals

2. Microwaves

Microwaves are high-frequency radio waves with shorter wavelength. They carry more data but travel in straight lines.

They are widely used in point-to-point communication because they can carry large amounts of data at high speed.



How they work

- Sent from one antenna tower to another
- Requires line of sight (no tall buildings or mountains in between)

Uses

- Mobile networks (4G, 5G calls and internet)
- Satellite communication
- Point-to-point communication (tower to tower)

Advantages

- High data transmission rate
- Suitable for long-distance communication

Limitations

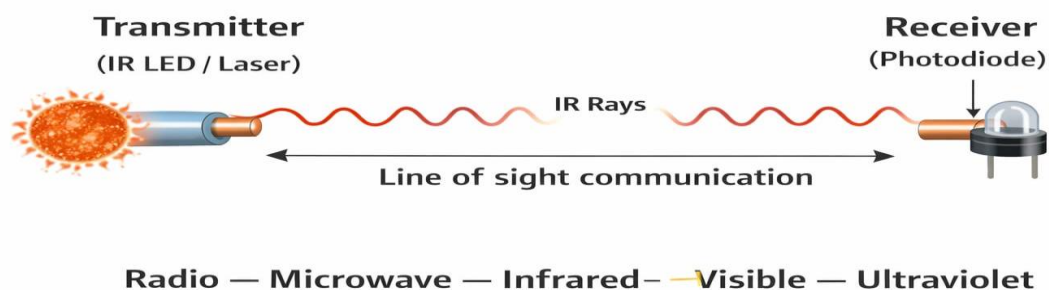
- Cannot bend around obstacles
- Affected by rain and weather
- Needs accurate antenna alignment

3. Infrared

Infrared is very short-range electromagnetic radiation. It works only over small distances.

Infrared (IR) communication is a type of unguided transmission that uses infrared light waves (just below the visible spectrum) to transmit data over short distances. It is commonly used in remote controls, wireless devices, and short-range communication systems.

Infrared communication is the transmission of data using infrared light waves (wavelength 700 nm – 1 mm) through free space, typically over short distances and line-of-sight paths.



How it works

- Sender emits infrared light
- Receiver detects the light signal
- Needs line of sight

Uses

- TV and AC remote controls
- Short-range device communication
- Sensors (motion detection)

Advantages

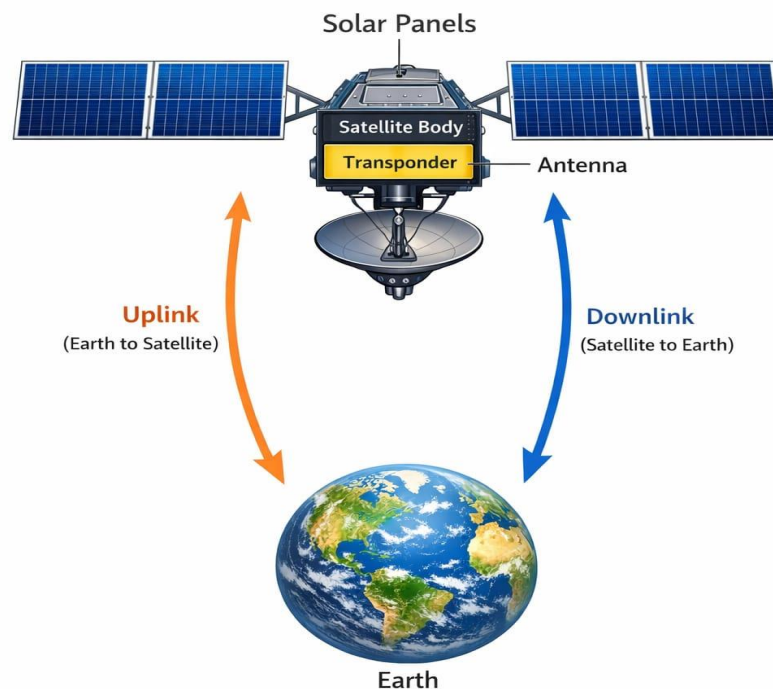
- Simple and low cost
- No radio interference

Limitations

- Very short distance
- Cannot pass through walls
- Needs direct alignment

4. Satellite Communication

It uses artificial satellites orbiting the Earth to relay signals between distant locations.



How it works

1. Earth station sends signal to satellite
2. Satellite amplifies and retransmits it
3. Signal reaches another Earth station

Uses

- GPS navigation
- TV broadcasting (DTH)
- Weather forecasting

- Global internet & phone services

Advantages

- Very wide coverage (global)
- Works in remote areas
- Reliable for long distances

Limitations

- High cost
- Signal delay (latency)
- Weather can affect signal quality

Analog Transmission-Modem

Analog transmission is the process of sending information in the form of continuous signals that vary in amplitude, frequency, or phase.

Explanation

- Signals are continuous waves
- More affected by noise and interference
- Used in radio, television, and traditional telephone systems

Role of Modem in Analog Transmission

Yes, modem is used when digital devices communicate over analog channels.

- Computer data → digital
- Telephone line → analog
- Modem converts digital data into analog signal and back

Modem in Analog Transmission

In analog transmission, a modem is a device that allows digital data to be transmitted over an analog communication channel.

Working

1. **Modulator**
 - Converts digital data into analog signal
2. **Transmission**
 - Analog signal travels through analog medium
3. **Demodulator**
 - Converts analog signal back into digital data

Modem

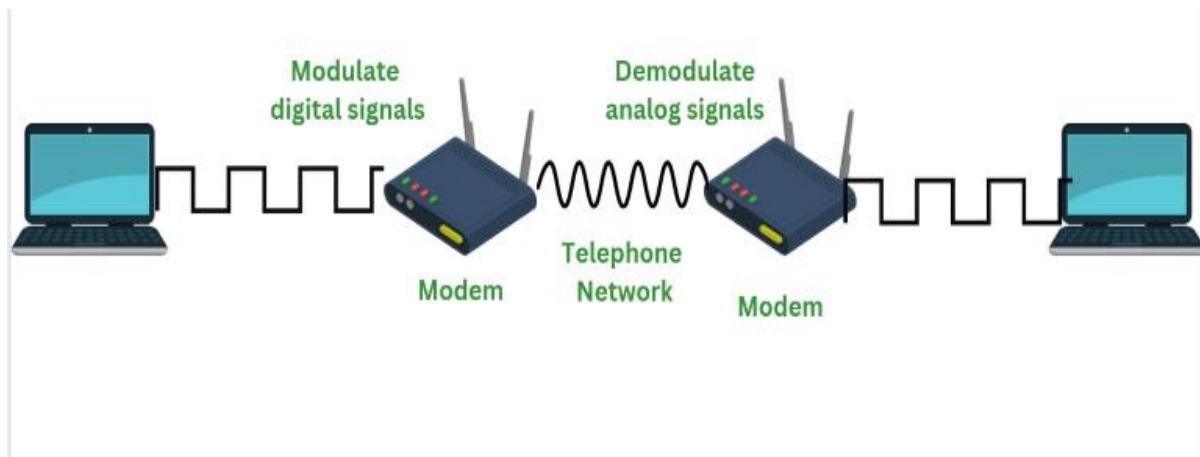
A Modem is a device that converts digital data into analog signals for transmission and converts analog signals back into digital data at the receiving end. Modem stands for Modulator and Demodulator.

Role of Modem in Digital Transmission

- Sender side: Digital data → Analog signal (Modulation)
- Receiver side: Analog signal → Digital data (Demodulation)

Note:

- Digital transmission uses 0s and 1s
- More reliable than analog transmission
- Modem is required for digital-to-analog conversion
- Widely used in networking and internet



Modulator

A Modulator is a device or process that converts digital data into an analog signal suitable for transmission over a communication channel.

Function of Modulator

- Takes digital data (0s and 1s)
- Converts it into analog signal
- Used at the sender side

Example

When a computer sends data through a telephone line, the modulator converts digital data into analog form.

Demodulator

A Demodulator is a device or process that converts a received analog signal back into digital data.

Function of Demodulator

- Takes analog signal
- Converts it into digital data (0s and 1s)
- Used at the receiver side

Example

At the receiving end, the demodulator converts the incoming analog signal into digital data that the computer can understand.

Digital transmission means sending data in digital form (0s and 1s) over a communication channel.

Digital transmission is the process of sending data in digital form (0s and 1s) from one device to another through a communication channel such as cable, fiber, or wireless medium.

Explanation

- Data is represented as **binary digits (0 and 1)**
- Signals are less affected by noise
- Used in computers, mobile communication, and internet

To transmit digital data, we use encoding techniques.

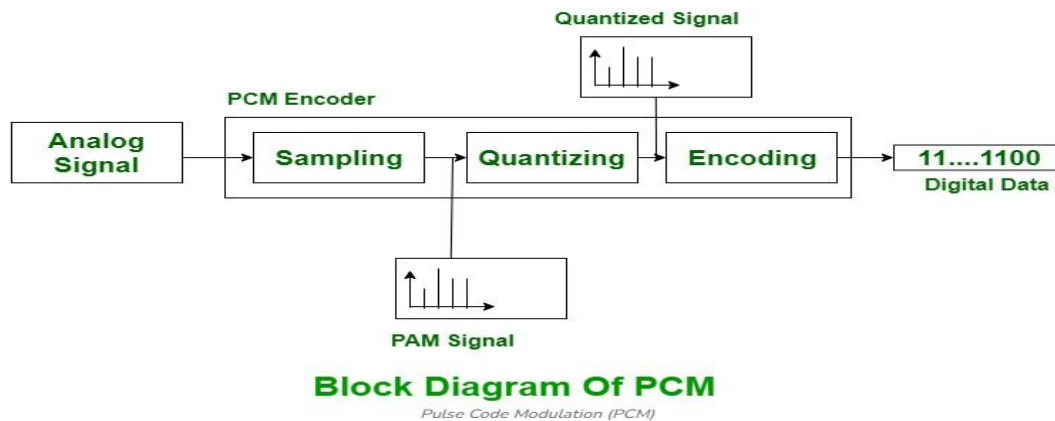
1. Pulse Code Modulation (PCM)
2. Manchester Coding
3. Differential Manchester Coding

1. Pulse Code Modulation (PCM)

Digital transmission is the process of sending data in the form of discrete signals (0s and 1s) over a transmission medium. It is widely used in modern communication systems due to its higher noise immunity and better data integrity compared to analog transmission.

Pulse Code Modulation (PCM) is a method of digital transmission in which analog signals are converted into digital signals by sampling, quantization, and encoding.

It is mainly used for voice transmission in telephone systems.



Steps in PCM

1. Sampling

- Analog signal is measured at regular time intervals
- Higher sampling rate = better quality

Sampling is the first step in PCM. In this step, the analog signal is taken at equal time intervals to convert it into discrete values.

There are three types of sampling:

1. Ideal Sampling

The signal is sampled using very narrow pulses. This method is theoretical and not used in practice.

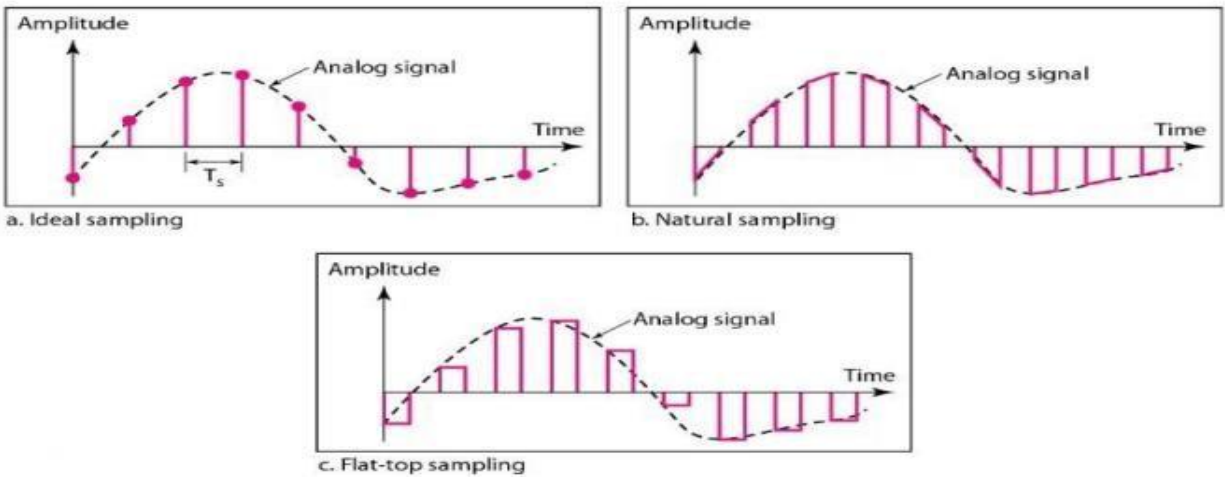
2. Natural Sampling

The signal is sampled for a short time using a fast switch. The samples follow the shape of the original signal.

3. Flat-top Sampling (Sample and Hold)

This is the most commonly used method. Each sample is held constant for a short time, giving flat-top pulses.

Sampling is also called **Pulse Amplitude Modulation (PAM)**.

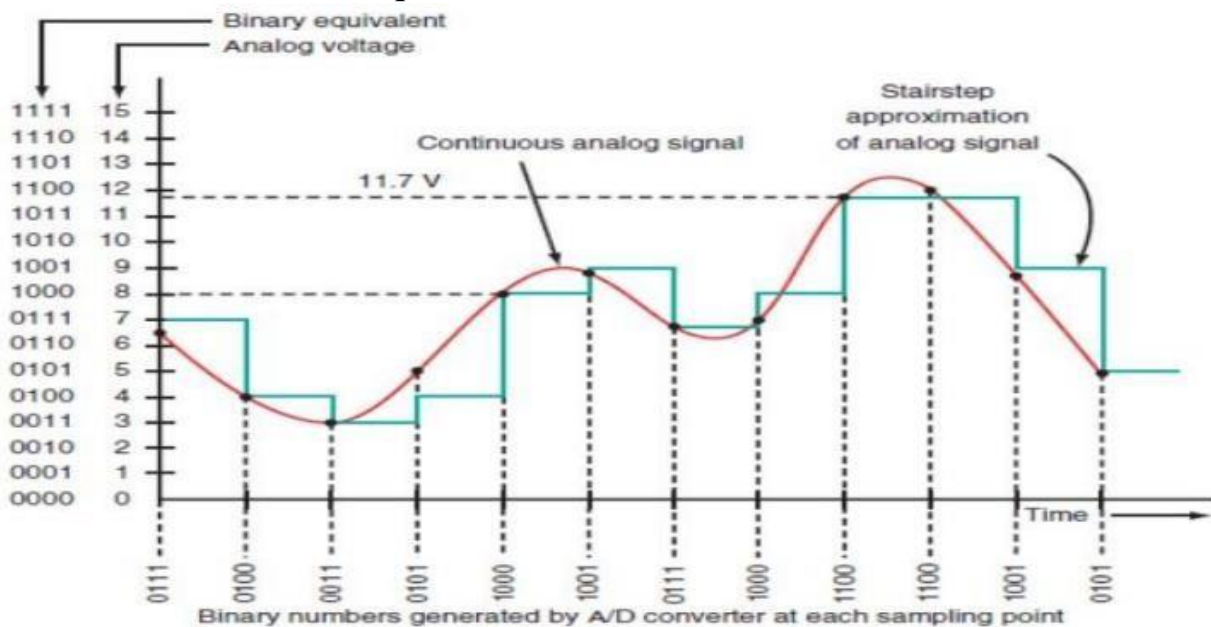


2. Quantization

- Sample values are rounded off to fixed levels

Quantization is the process of converting sampled signal values into a fixed number of levels.

- The sampled values lie between **V_{min}** and **V_{max}**.
- These values are rounded off to the nearest fixed level.
- This makes the signal suitable for digital processing.
- A small error called **quantization error** is introduced.



3. Encoding

- Quantized values are converted into binary (0s and 1s)

Encoding is the final step in PCM. In this process, each quantized sample is represented by a binary code word consisting of a fixed number of bits.

- The number of bits per sample depends on the number of quantization levels.
- Each quantized level is assigned a unique binary code.

For example:

- Quantization level 2 $\rightarrow$ 010
- Quantization level 5 $\rightarrow$ 101

2: Manchester Coding

Manchester coding is a way to send data where every bit (0 or 1) changes its signal in the middle.

This middle change helps the receiver know the timing (clock).

Line coding is a method of converting digital data (0s and 1s) into digital signals suitable for transmission over a physical medium.

Manchester coding and Differential Manchester coding are two widely used self-clocking line codes. They combine data and clock information in the signal.

Every bit has a middle transition

- 1 $\rightarrow$ signal goes LOW to HIGH
- 0 $\rightarrow$ signal goes HIGH to LOW

Bit = 1
 LOW | HIGH
 _____ | _____
 |
 ↑
 middle change

Bit = 0
 HIGH | LOW
 _____ | _____
 |
 ↑
 middle change

Binary data: 1 0

For 1

- First half $\rightarrow$ LOW
- Middle $\rightarrow$ change
- Second half $\rightarrow$ HIGH

For 0

- First half → HIGH
- Middle → change
- Second half → LOW

Why Manchester Coding is Used?

- Receiver can easily follow timing
- No confusion even with many 0s or 1s
- Used in Ethernet

Manchester coding is a digital line coding technique in which each bit has a transition in the middle of the bit period.

What is Manchester Encoding?

- In Manchester encoding, a low-to-high transition represents a 1, and a high-to-low represents a 0.
- The duration of the bit is divided into two levels.
- Voltage remains at one level in one half and moves to another level in the second half.

NOTE :

+ve voltage = 1

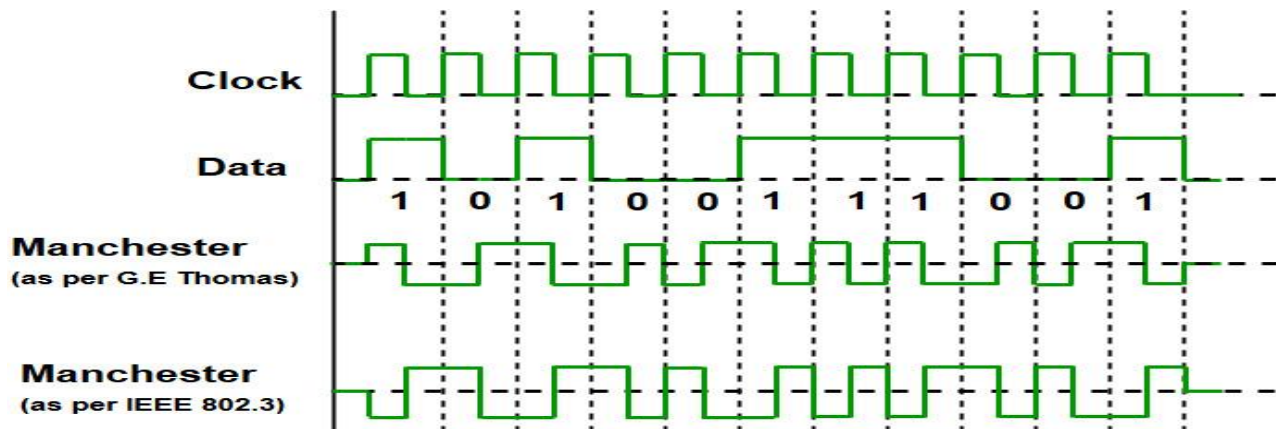
-ve voltage = 0

There are two types of conventions in Manchester encoding:

1. Dr. Thomas: In this manchester encoding 0 is represented as low-to-high and 1 is represented as high-to-low.

Dr. Thomas Manchester Encoding

2. IEEE802.3: In this manchester encoding, 0 is represented as high-to-low and 1 is represented as low-to-high.



Manchester coding is a line coding technique used to transmit digital data

3. Differential Manchester Coding

Differential Manchester Coding is a digital line coding technique in which data bits are represented by changes (transitions) in the signal, not by voltage level itself.

- A transition in the middle of every bit time is compulsory (for clock synchronization).
- The presence or absence of a transition at the beginning of the bit period decides the data value.

Bit Representation Rule

- Bit 0 → Transition at the beginning of the bit period
- Bit 1 → No transition at the beginning of the bit period
- Middle transition → Always present (for clock)

Data depends on change, not on high or low level.

Example

Data bits:

1 0 1 1 0

Explanation:

- For 1 → no transition at start, only middle transition
- For 0 → transition at start + middle transition

(Exact waveform depends on previous signal, which is why it's called *differential*)

It is an improved version of Manchester coding.

Here, data is represented by transition at the beginning of the bit period.

How it works

- 0 → Transition at the beginning of bit
- 1 → No transition at the beginning
- Middle transition always occurs (for clocking)

Features

- Less sensitive to polarity inversion
- Reliable synchronization

Advantages

- Better noise immunity
- Polarity-independent

Disadvantages

- Complex implementation
- Requires more bandwidth

Differential Manchester Coding

Definition

Digital line coding technique using **changes (transitions)** in the signal, not voltage levels.

Bit Representation Rule:

- **Transition** in the middle of every bit
- **Start transition** determines the data bit.

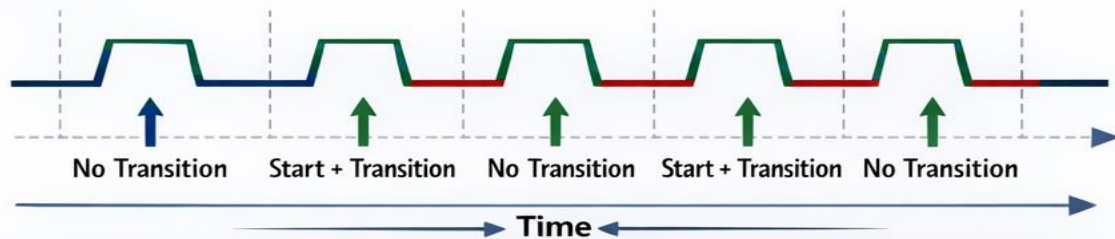
↑ **Bit 0:** Transition at the **Beginning**

↓ **Bit 1:** No Transition at the **Beginning**

Middle transition: Always present (for clock).

Example: Data: 1 0 1 1 0

Bits:



Advantages:

- ✓ Clock Synchronization
- ✓ Polarity Independent
- ✓ Reliable Signal

Disadvantages:

- ✗ More Bandwidth
- ✗ More Complex

Applications:

- Token Ring Networks
- Digital Communication Systems

Memory Tip: “Differential Manchester = Data by **transition**, not by level.”

Modulation: -

Modulation is the process of **modifying a carrier signal** (it is a high-frequency wave that carries the message(information) signal from transmitter to receiver.) to **carry information** from a message or data signal.

- Carrier signal: High-frequency signal used to transmit data.
- Message signal: The original signal (analog or digital) to be transmitted.

Purpose of Modulation

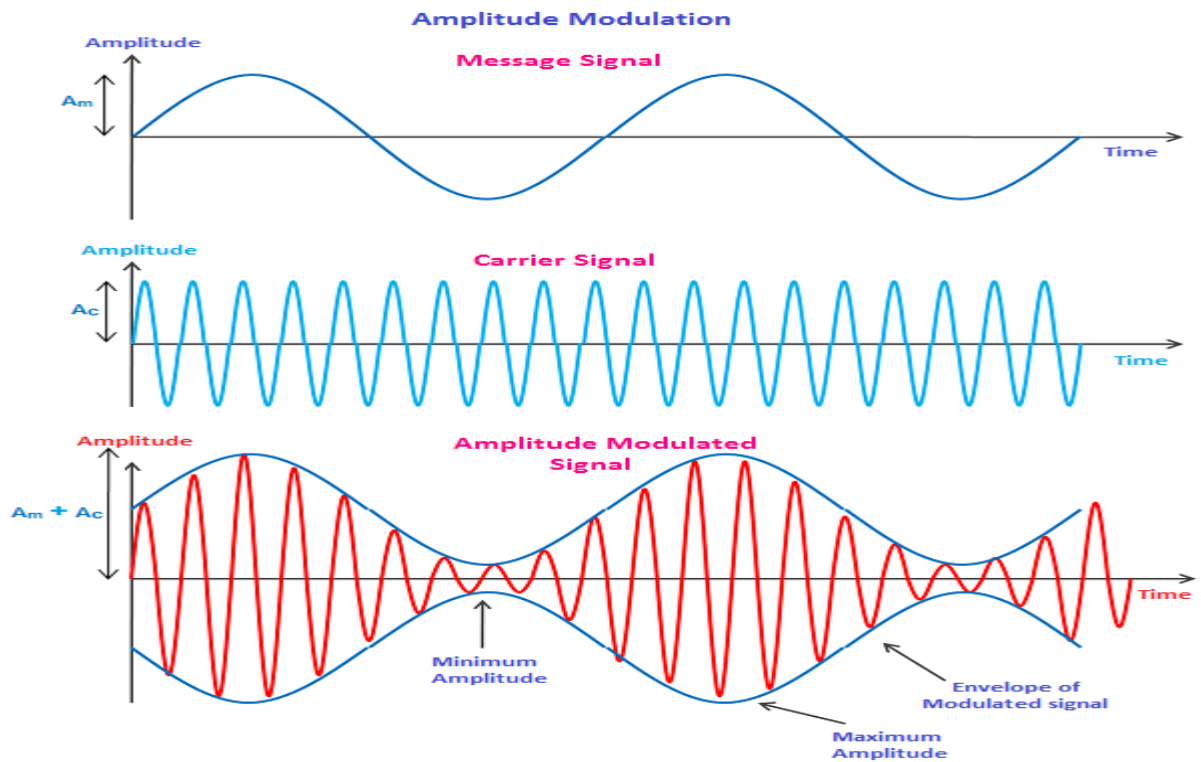
1. **Increase transmission range** – Converts low-frequency signals to high-frequency carriers so they can travel long distances.
2. **Reduce antenna size** – High-frequency carriers have shorter wavelengths, allowing practical antenna sizes.
3. **Prevent mixing of signals** – Assigns each signal a unique carrier frequency to avoid interference.
4. **Improve signal quality** – Certain modulation methods reduce noise and distortion during transmission.
5. **Enable multiplexing** – Allows multiple signals to share the same channel by using different carrier frequencies.

Types of Modulation

Modulation is mainly classified into **three types** based on which property of the carrier is varied:

1. Amplitude Modulation (AM)

- **Definition:** The **amplitude of the carrier signal** is varied according to the Real-time amplitude of the message signal, while frequency and phase remain constant.



- Only the **height** changes.

Characteristics

- Amplitude of carrier changes
- Frequency and phase remain constant
- Highly sensitive to noise
- Small bandwidth
- Low power efficiency
- Simple transmitter and receiver

Bandwidth

- Bandwidth = Highest frequency – Lowest frequency.

$$BW = 2f_m$$

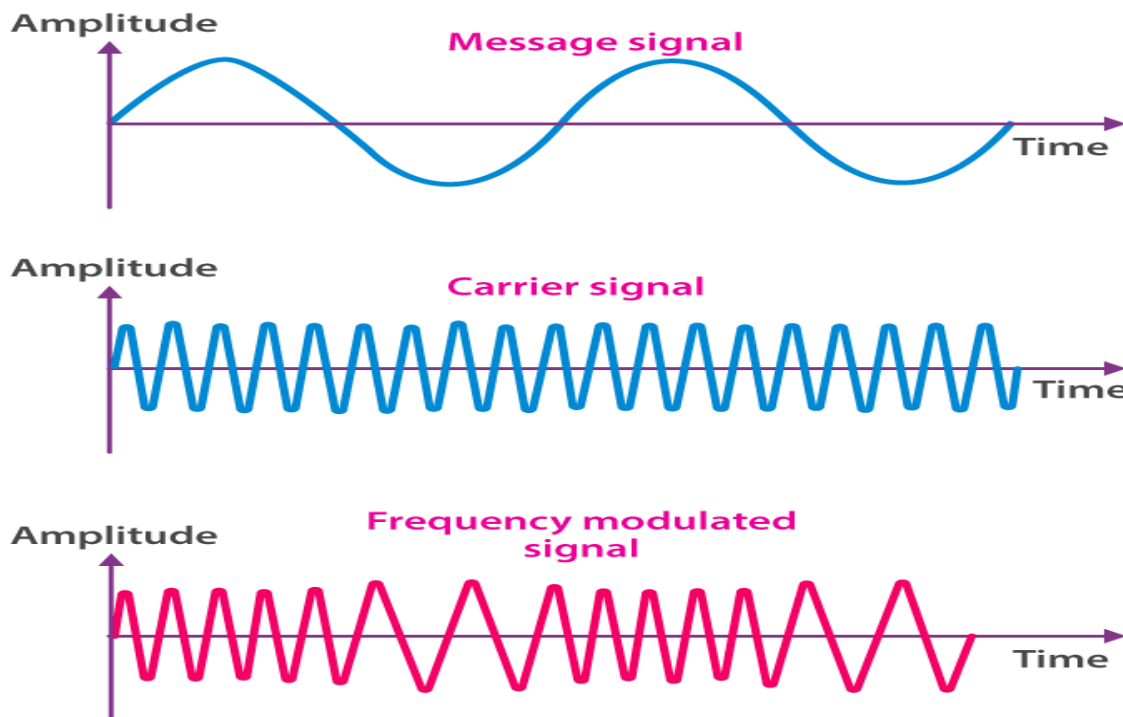
(where f_m = maximum message frequency)

Applications

- AM radio broadcasting
- Aviation communication
- Analog TV (older systems)

2. Frequency Modulation (FM)

- **Definition:** The **frequency of the carrier signal** is varied according to the Real-time amplitude of the message signal, while amplitude remains constant.



Only the **number of cycles per second** changes.

Characteristics

- Frequency of carrier changes
- Amplitude remains constant
- Less sensitive to noise than AM
- Better sound quality
- Higher bandwidth than AM
- Constant power transmission

Bandwidth (Carson's Rule)

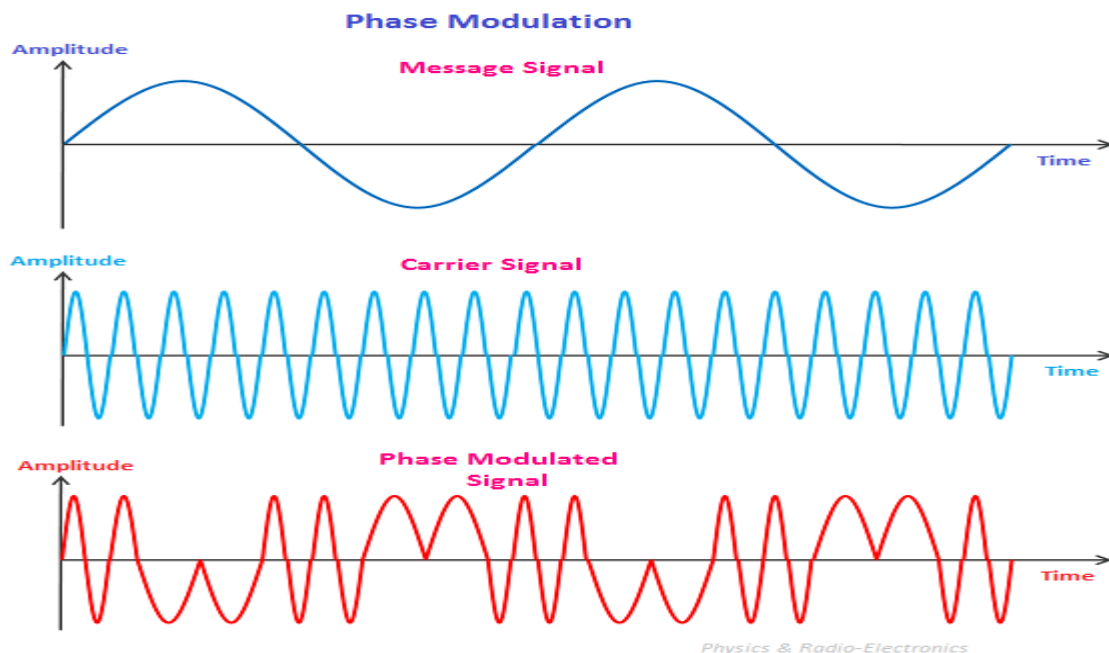
$$BW = 2(\Delta f + f_m)$$

Applications

- FM radio broadcasting
- TV sound transmission
- Two-way radio communication

3. Phase Modulation (PM)

- **Definition:** The **phase of the carrier signal** is varied according to the Realtime amplitude of the message signal, while amplitude and frequency remain constant.
-



- The wave is **shifted left or right**, but height and spacing stay the same.

Characteristics

- Less sensitive to noise than AM
- Bandwidth depends on amplitude and rate of the message signal
- Used in digital modulation schemes
- Phase of carrier changes according to message signal
- Amplitude remains constant

Applications

- Used as part of FM systems (FM can be generated indirectly using PM)
- Used in digital modulation techniques like:

- PSK (Phase Shift Keying)
- QPSK (Quadrature Phase Shift Keying)
- Used in modern communication systems

Transmission modes:-

It refers to the method or manner in which data is transmitted from one device to another in a communication system.

Transmission modes define how data flows between sender and receiver, including the direction and timing of data transfer.

It explains:

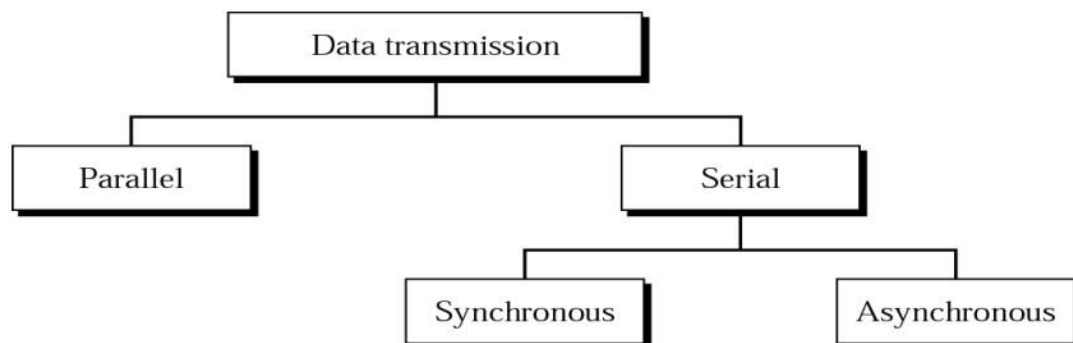
- How data is sent
- How many bits are sent at a time
- How sender and receiver stay synchronized

Types of Transmission Modes

Transmission modes are commonly classified as:

1. Parallel Transmission – Multiple bits are sent at the same time
2. Serial Transmission – Bits are sent one after another
3. Synchronous Transmission – Data is sent using a common clock
4. Asynchronous Transmission – Data is sent with start and stop bits

Fig. Data transmission



Parallel transmission:-

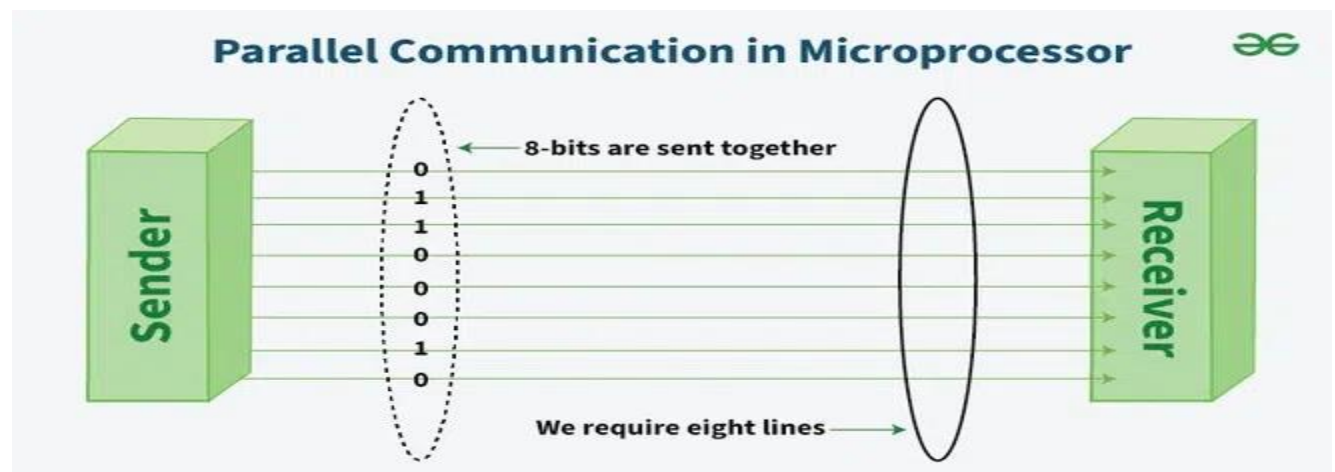
It is a transmission mode in which multiple bits of data are sent simultaneously, with each bit traveling on a separate communication line.

This method allows faster data transfer since many bits are transmitted at the same time.

However, it is mainly used for short-distance communication because signal interference and synchronization issues increase over long distances.

Characteristics

- Multiple data lines are used
- Several bits (usually 8 or more) are transmitted at the same time
- Requires synchronization between all lines
- Suitable for short-distance communication



Advantages

- Very high data transfer speed
- Faster than serial transmission over short distances

Disadvantages

- Requires more wires, increasing cost
- Not suitable for long-distance transmission

Serial transmission:-

There are two types of Serial transmission:

a) Synchronous transmission:

b) Asynchronous transmission:-

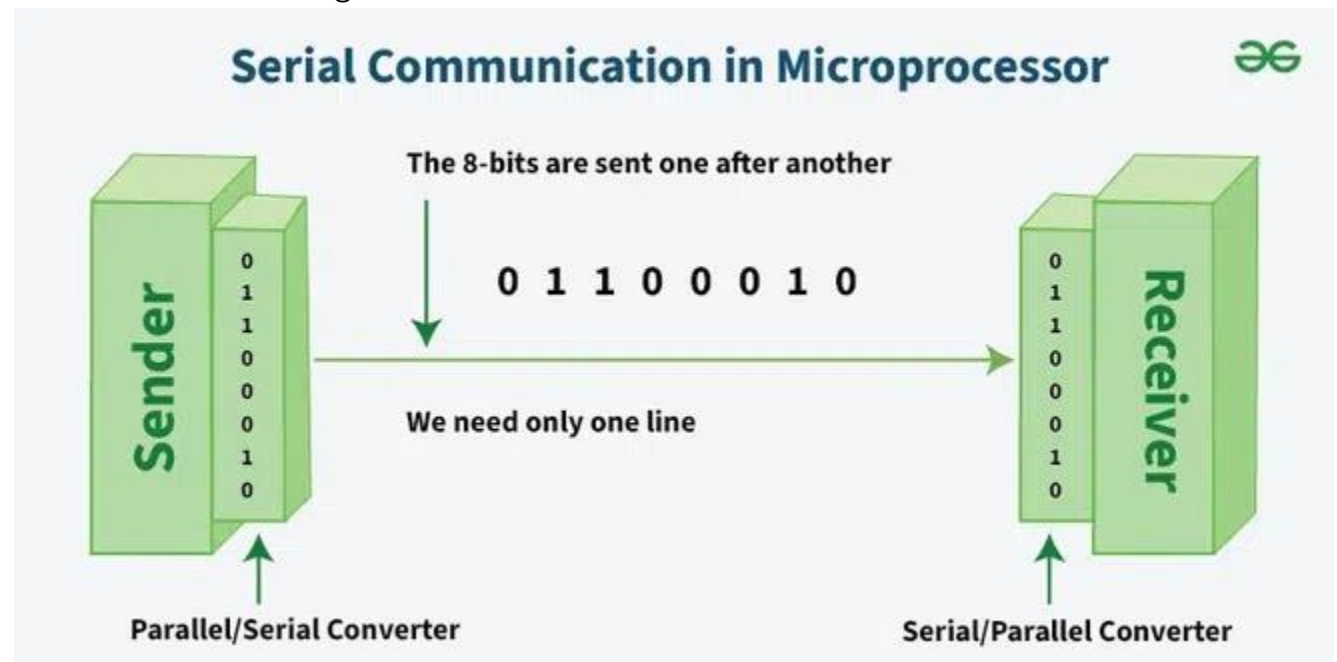
It is a transmission mode in which data is sent one bit at a time, sequentially, over a single communication line.

This method is more reliable for long-distance communication since it reduces interference between signals.

Serial transmission is widely used in modern communication systems.

Characteristics

- Uses a single data line
- Bits are transmitted one after another
- Can operate in synchronous or asynchronous mode
- Suitable for long-distance communication



Advantages

- Requires fewer wires, reducing cost
- Less signal interference and noise
- Reliable over long distances

Disadvantages

- Slower than parallel transmission for short distances
- Data is transmitted sequentially, which may reduce speed

a)Synchronous transmission:

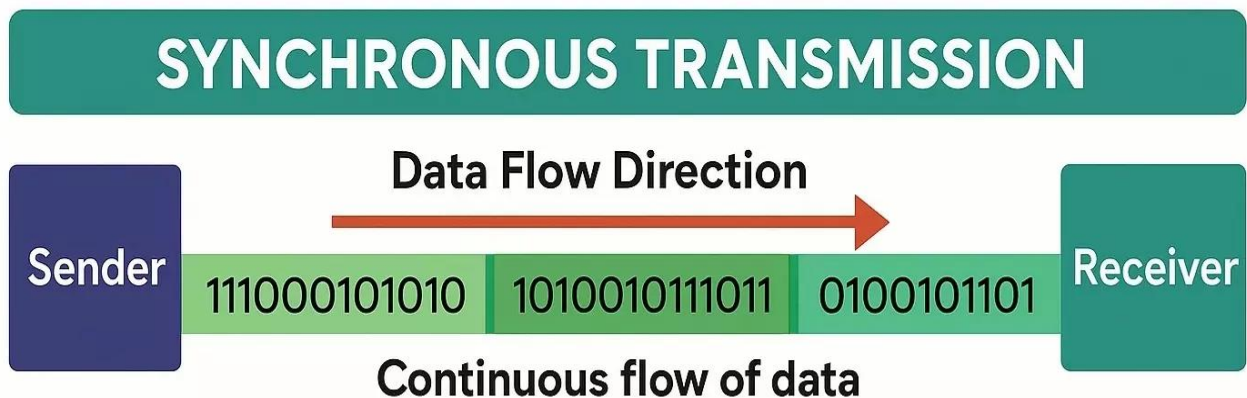
Synchronous transmission is a transmission mode in which data is sent continuously in blocks or frames, synchronized by a common clock signal shared between the sender and receiver.

There are no start and stop bits for each character, which makes data transfer more efficient.

It is commonly used for high-speed data communication.

Characteristics

- Uses a common clock for synchronization
- Data is transmitted in continuous streams or frames
- No start and stop bits for individual characters
- High data transmission speed



Advantages

- Efficient use of bandwidth
- Higher data transfer rate
- Suitable for large amounts of data

Disadvantages

- Requires complex hardware
- Accurate clock synchronization is necessary

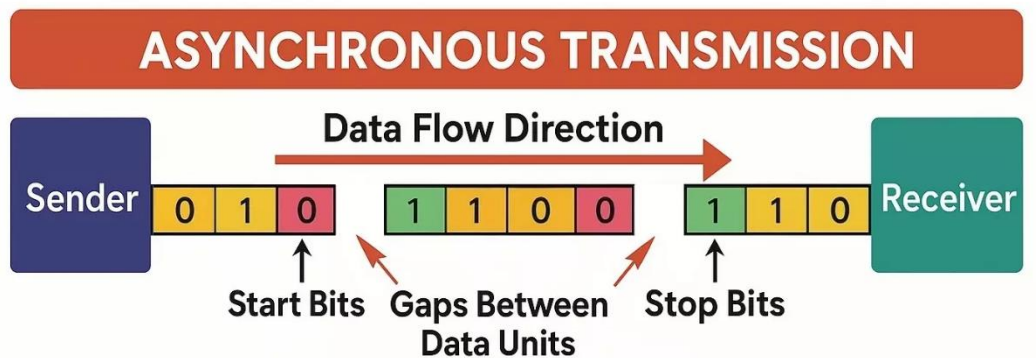
b) Asynchronous transmission:-

Asynchronous transmission is a transmission mode in which data is sent one character at a time without using a common clock signal between the sender and receiver.

Each character is framed with start and stop bits to maintain synchronization. It is commonly used for low-speed and simple communication.

Characteristics

- No shared clock between sender and receiver
- Each character has start and stop bits
- Data is transmitted intermittently
- Simple and flexible transmission method



Advantages

- Simple and low-cost implementation
- No need for clock synchronization
- Suitable for low-speed data transmission

Disadvantages

- Lower efficiency due to start and stop bits

Slower than synchronous transmission

Multiplexing:

Multiplexing is like **sharing one road for many cars**.

- Imagine you have **one wire** or **one fiber**.
- You want **many signals** (like phone calls, internet data, or TV channels) to travel at the same time.
- Multiplexing **combines these signals** so they can all travel together **without crashing into each other**.

So, instead of needing a separate wire for every signal, you can send many signals **on the same wire**.

Multiplexing is a technique used in communications to **combine multiple signals and send them over a single communication channel** simultaneously.

Think of it like a multi-lane highway: instead of building separate roads for each car, multiplexing lets many cars (signals) share the same road (channel) without colliding, by separating them in **frequency, time, or wavelength**.

Key points:

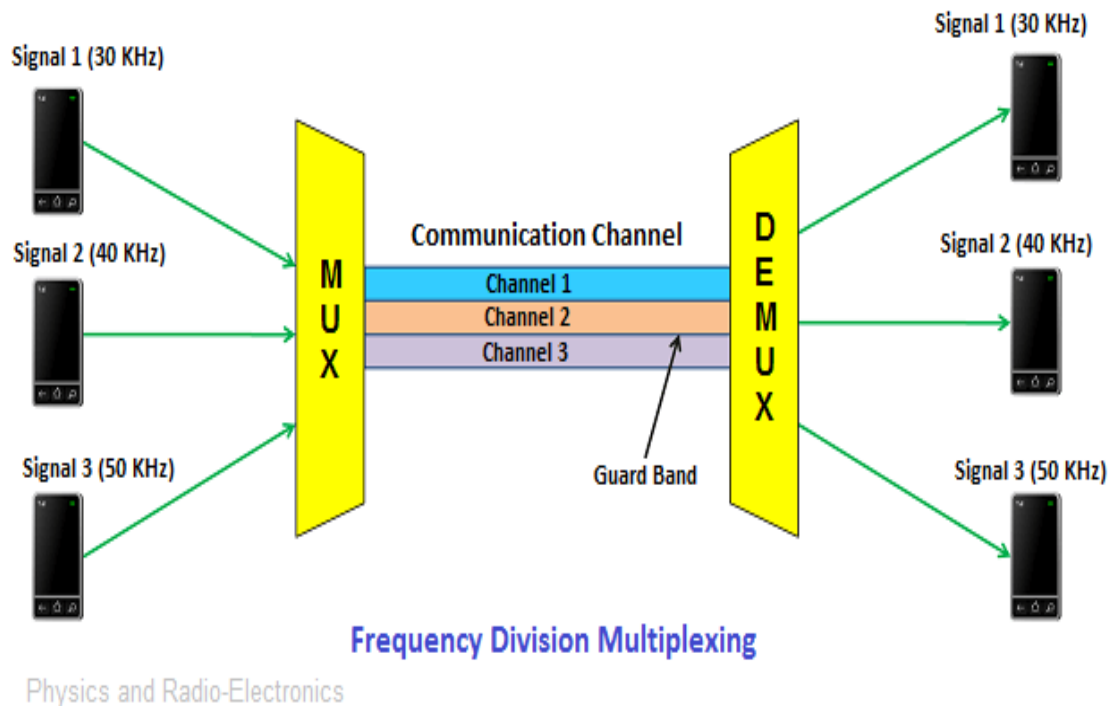
- **Purpose:** Efficiently use the available bandwidth and reduce the cost of transmission.
- **Applications:** Telephone networks, TV broadcasting, internet data over fiber optics, satellite communication.

In simple terms: Multiplexing allows multiple conversations, data streams, or signals to travel together over the same medium without interfering with each other.



1-Frequency Division Multiplexing (FDM)

- Each signal gets its own **frequency** to travel on the same channel.
- Think of it like **radio stations**: each station has a different frequency, so you can listen to many stations without them mixing up.



How it works:

1. Suppose you have 3 signals: A, B, and C.
2. FDM assigns:
 - Signal A → 100–110 MHz
 - Signal B → 110–120 MHz
 - Signal C → 120–130 MHz
3. All three travel **together on the same medium**, but at **different frequencies**.
4. At the receiver, they are separated by frequency filters.

Real-life examples:

- Radio stations: Each station uses a different frequency, so you can listen to 100.1 FM or 101.5 FM without interference.
- Cable TV: Many TV channels are sent through one cable, each on a different frequency.
- Older telephone systems: Multiple phone calls over the same wire using different frequency bands.

2-Time Division Multiplexing (TDM) –

TDM is a way to send **many signals over the same channel** by giving each signal its **own time slot**.

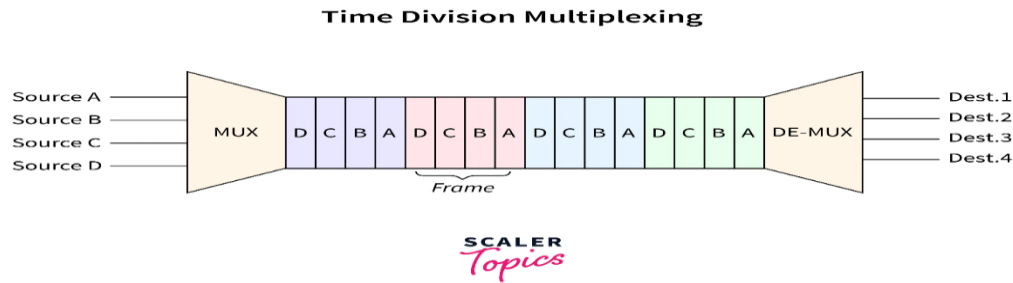
- Think of it as **sharing one lane on a highway**, but taking turns instead of having separate lanes.
- Only **one signal travels at a time**, but they switch so fast that it seems like all signals are traveling together.

How it works

1. Suppose you have 3 signals: A, B, C.
2. TDM assigns **time slots**:
 - Time slot 1 → Signal A
 - Time slot 2 → Signal B
 - Time slot 3 → Signal C
3. The channel sends the signals **one after another** in these time slots.
4. At the receiver, the signals are **reassembled** in the same order.

Real-life examples:

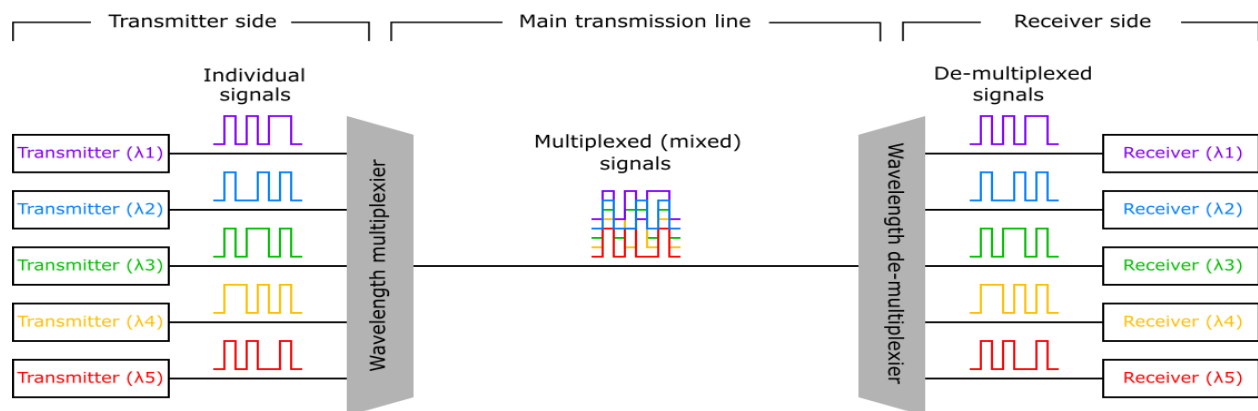
- **Digital telephony (like ISDN)**: Multiple calls over the same line using time slots.
- **Computer networks**: Sharing a network connection among multiple devices.
- **Satellite communication**: Many signals share the same channel by taking turns.
- Imagine **one-lane bridge** with 3 cars waiting.
- Car A goes first (time slot 1), then Car B (time slot 2), then Car C (time slot 3).
- If they move very fast, it looks like all cars are crossing simultaneously.



3-Wavelength Division Multiplexing (WDM) –

WDM is a technique used in **optical fiber networks** to send **many signals at the same time**, but each signal travels on a **different wavelength (color) of light**.

- Think of it like **sending multiple laser beams through the same fiber**, each a different color.
- At the receiver, the colors are **separated** to get the original signals.



How it works

1. Suppose you have 3 signals: A, B, C.
2. Each signal is converted to light of a different **wavelength (color)**:
 - Signal A → Red light (λ_1)
 - Signal B → Green light (λ_2)
 - Signal C → Blue light (λ_3)
3. All three light signals are **combined** and sent through the same optical fiber.
4. At the receiving end, a device called a **demultiplexer** separates the signals by their wavelengths.

Real-life examples:

- High-speed internet over fiber optics
- Telecommunication backbone networks
- Long-distance data transmission.

Switching:-

Switching is the process of forwarding data from one device to another in a network using intermediate devices like switches or routers.

When a device sends data, switching decides **which path the data should take** to reach the correct destination.

Example:

Suppose **Computer A** wants to send a file to **Computer B** in a college lab network.

Computer A → Switch → Computer B

Computer A sends the data.

The **network switch** checks the destination address (MAC address).

The switch looks in its MAC address table.

It forwards the data only to the port where Computer B is connected.

So, the switch **decides the correct path (port)** and sends the data only to the intended device — not to all computers.

How Does a Network Switch Works?



1.Circuit Switching

Circuit Switching is a switching technique in which a **dedicated communication path** is established between sender and receiver before data transfer begins.

Once the connection is set up, the entire communication uses that same path until it ends. Circuit switching is a technique where a dedicated path is established between sender and receiver for the entire duration of communication.

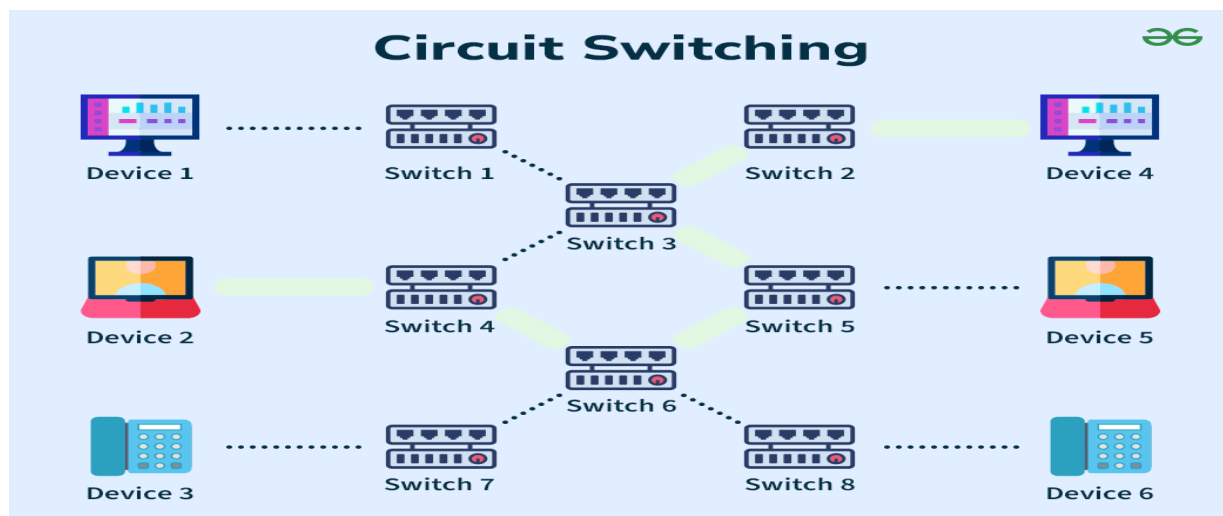
Before Communication:

No connection

After Connection Setup:

Sender ————— Dedicated Path ————— Receiver

The path remains reserved and cannot be used by others during communication.



Phases of Circuit Switching

Setup Phase – Connection is established

Data Transfer Phase – Data is transmitted

Termination Phase – Connection is disconnected

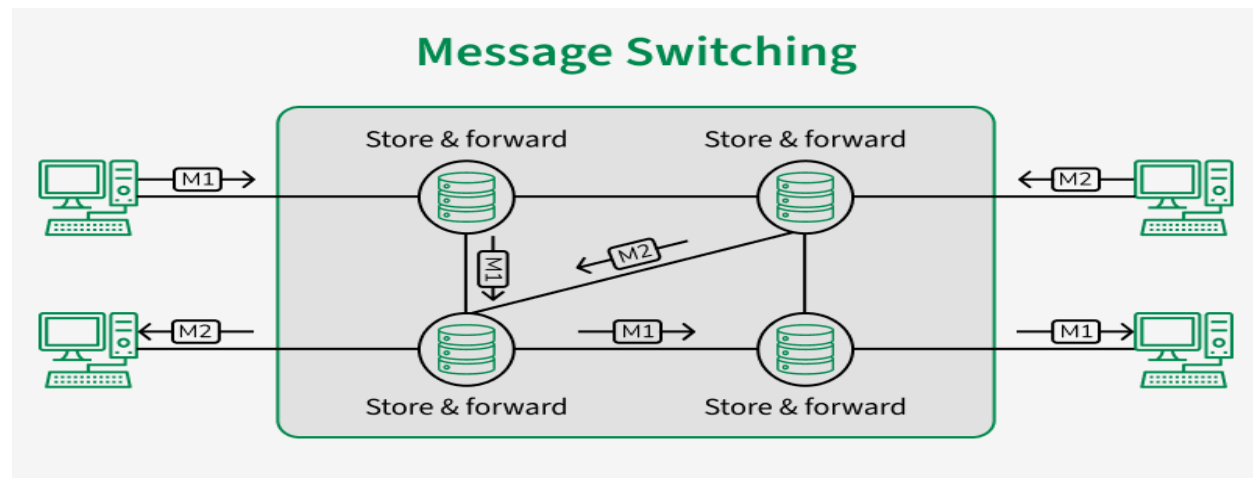
Advantages

- Reliable communication
- No delay after connection setup
- Fixed bandwidth

Disadvantages

- Wastes bandwidth when no data is sent
- Setup time required

2.Message Switching



Message switching is a technique in which the entire message is sent to an intermediate node, stored there temporarily, and then forwarded to the next node until it reaches the destination. No dedicated path is required.

How It Works

1. **Message Creation:** The sender creates the complete message.
2. **Store-and-Forward:** The first intermediate node receives the full message and stores it temporarily.
3. **Forwarding:** The node forwards the message to the next node when the path is free.
4. **Delivery:** The message continues through nodes until it reaches the destination.

Diagram:

Sender → [Node A] → [Node B] → Receiver
(Store & Forward) (Store & Forward)

Example

- Sending an email via multiple mail servers
- Telegrams or postal message systems

The message may be stored in servers if the receiver is busy or offline.

Advantages

- No dedicated path needed
- Can store messages if the next node or destination is busy
- Efficient use of network resources

Disadvantages

- High delay due to storage at each node
- Requires large storage capacity at intermediate nodes
- Slower compared to packet switching

3. Packet Switching

Packet switching is a technique in which data is divided into small units called **packets**, which are sent independently over a network and reassembled at the destination. Each packet contains part of the data along with source and destination information.

How It Works

1. **Data Division:** The sender divides the message into small packets.
2. **Addressing:** Each packet is labeled with the destination address.
3. **Routing:** Packets travel independently through the network, possibly taking different paths.
4. **Reassembly:** At the destination, packets are reassembled in the correct order to form the original message.

Diagram:

Sender → Packet 1 → Router A → Receiver
→ Packet 2 → Router B → Receiver
→ Packet 3 → Router C → Receiver

Example

- Sending an email
- Accessing websites on the Internet
- Online messaging apps

All these use packet switching to efficiently transmit data across networks.

Advantages

- Efficient use of network bandwidth
- Multiple users can share the same network
- Reliable delivery with error checking
- Scalable for large networks

Disadvantages

- Packets may arrive out of order
- Variable delay (latency) can occur
- Requires more complex protocols to reassemble packets

